

AI INVESTMENT STRATEGIST

Technical Specification Document & System Architecture v1.0

1. Project Overview

A high-performance financial analysis platform that combines real-time market data with local Large Language Models (LLMs) to provide automated portfolio rebalancing advice and risk assessment.

2. Technical Stack

Backend

FastAPI (Python)

Database

MongoDB

AI Engine

Ollama (Llama 3)

Market Data

yfinance API

3. Data Architecture

3.1 Portfolio Collection

Stores user metadata and risk preferences.

```
{
  "_id": ObjectId,
  "user_id": String,
  "portfolio_name": String,
  "risk_profile": "Conservative" | "Balanced" | "Aggressive",
  "base_currency": "USD"
}
```

3.2 Holdings Collection

Tracks specific assets within a portfolio.

```
{
  "portfolio_id": ObjectId,
  "ticker": String,
  "shares": Float,
  "average_buy_price": Float,
  "sector": String,
  "asset_class": "Equity" | "Crypto" | "Fixed Income"
}
```

4. Core System Logic

4.1 Live Market Synchronization

The system utilizes `yfinance` to fetch current pricing and calculate Unrealized Profit/Loss (uP&L) without storing expensive historical data locally.

4.2 AI Strategy Generator

The AI Engine receives a JSON-structured snapshot of the portfolio and generates advice based on the `risk_profile`. Key prompts focus on sector concentration and rebalancing targets.

Key Metric: Diversity Score

The backend calculates the percentage concentration per sector. If any sector (e.g., Technology) exceeds 40% of the total value, the AI is triggered to suggest specific sell-offs.

5. Development Roadmap

- **Phase 1:** Implement MongoDB CRUD for portfolio and asset tracking.
- **Phase 2:** Build the Price Sync engine using `yfinance` to calculate real-time portfolio value.

- **Phase 3:** Integrate Ollama local API to provide natural language critiques of holdings.
- **Phase 4:** Develop the frontend dashboard with Dark/Light mode toggle and Chart.js visuals.